

$$\begin{aligned} &= 1.225 - 1 \\ &= 0.225 \text{ (or 22.5\%)} \end{aligned}$$

So, the calculations based on Example 3 would give 22.5% CAGR returns in our case.

This method of compounded return calculation is more realistic and represents how the money "actually" grows in your saving banks or fixed deposits that give a definite interest rate for a defined period, assuming that you have not opted for interest pay out option. The interest pay out option essentially means that the annual return is paid out to you separately and the money does not grow in a compounded manner.

Note that in comparison to Example 2, which used simple rate of interest, you could get the same returns in Example 3 using a lesser rate of return (22.5% with CAGR as against 25% with Simple ROI). This goes to prove the importance of compounded interest. It also proves the fact that if the rate of interest is the same for Simple ROI and CAGR, then CAGR gives you higher returns. The CAGR formula is the one which can help you calculate the immense power of compounding over a period of time. We will see the power of compounding in more detail later in the book.

However, there is still a limitation with this formula. It assumes that you invest the money once and then let it grow in a compounded manner and get all the returns on a definite date. With traditional investment tools like savings accounts and fixed deposits, this formula works fairly accurately because these two types of investment work within the constraints of a definite amount invested for a definite period.

However, for your investments to really give you returns that beat inflation, we need to diversify our investments to other investment tools. If that is the case, then how do we calculate the returns for investments like mutual funds, stocks and other investments, which take periodic as well as irregularly sized investments over irregular periods of times? Over and above this, part of the investments may mature during the defined time period. Actually, there is no defined time period or amount to invest or for returns in such cases. Understandably, these calculations can get really complex and it is difficult to obtain accurate results through a simple